



SCIENTIFIC INSTRUMENTS

SI-7000

Tank Gauging System

Operational Manual

Part A

SCIENTIFIC INSTRUMENTS, INC.
4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
090-7000-2A Rev -

Record of Revisions						
Rev	RFEA #	Description	Rev by	Date	Appr by	Date
-		Initial Release	KN	8/21/19	KN	8/21/19

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1 INTRODUCTION

1.1 GENERAL DESCRIPTION

The Scientific Instruments, Inc. SI-7000 is a state of the art electro-mechanical system designed to measure liquid level, temperature, and density in cryogenic liquids, including liquid natural gas (LNG), propane, and butane. The system has been designed with an emphasis on reliability, modularity, and ease of maintenance and operation. With its computer-controlled operation, the SI-7000 gives operators access to critical data necessary for intelligent and safe operation of the plant.

In addition to providing an accurate measure of the liquid level in a tank, the SI-7000 provides for precise measurement of temperature and density throughout the tank by positioning a multi-sensor probe assembly at any height in the tank. This makes it possible to obtain a profile that gives an accurate representation of the current conditions in the tank. This information is crucial for the safe storage and handling of cryogenic liquids, since they are subject to layering, and over time, possible “rollover” conditions.

1.2 APPLICATION

A significant safety concern in the storage of cryogenic liquids such as LNG is a phenomenon known as “rollover.” If this occurs, pressures inside the storage tank may rise to excessive levels. The tanks are equipped with safety vent valves that are designed to keep the pressures from rising to levels that could cause structural damage. However, when these valves operate, gas is vented to the atmosphere at an uncontrolled rate, which is an additional safety concern.

Rollover occurs under certain conditions as stratified LNG comes to equilibrium. Stratification occurs when the product in the tank forms in layers with different densities and different temperatures. Sudden mixing of LNG in any storage tank occurs as two or more layer densities approach equality. Any heat trapped in the system is released rapidly during mixing, generating a vapor which may exceed the venting capability of the tank.

By periodically taking a profile of the tank, the Scientific Instruments SI-7000 can detect stratification and possible rollover conditions and generate an alarm signaling the need for operator intervention. Operators can then take preventative measures (such as mixing the liquid or moving it to another tank).

1.3 SYSTEM CONFIGURATION

The integrated touch screen LCD is the primary interface used for setting all system parameters. The system configuration may also be changed remotely via a Modbus serial communication interface.

2 SYSTEM SPECIFICATIONS

2.1 CERTIFICATION STANDARDS

IEC 60079-0:2011, IEC 60079-1:2014, IEC 60079-11:2011, EN 60079-0:2012/A11:2013, EN 60079-1:2014, EN 60079-11:2012, ISO 80079-36, ISO 80079-37, IEC/EN61010-1:2010

2.2 EQUIPMENT MARKINGS

Electrical Compartment: II (1) 2 G Ex db [ia IIC Ga] IIB T4 Gb ($-20^{\circ}\text{C} \leq T_a \leq +50^{\circ}\text{C}$)

Mechanical Compartment: II 1 G Ex h IIC T4 Ga ($-20^{\circ}\text{C} \leq T_a \leq +50^{\circ}\text{C}$)

Probe Assembly: II 1 G Ex ia IIC T4 Ga ($-200^{\circ}\text{C} \leq T_a \leq +50^{\circ}\text{C}$)

2.3 MEASUREMENT CAPABILITY

Level:

Range 56 m
Resolution: 1 mm
Accuracy: ± 2 mm

Temperature:

Range -200°C to $+50^{\circ}\text{C}$
Resolution: 0.01°C
Accuracy: $\pm 0.1^{\circ}\text{C}$

Density:

Range 400 to 1000 kg/m^3
Resolution: 0.01 kg/m^3
Accuracy: $\pm 0.5\text{ kg/m}^3$
Profile Points: 250+

Sensor Isolation: Internal I.S. Barriers

2.4 ELECTRICAL

Operational Voltage: 85-240 VAC
Certified Operational Voltage: 120-240 VAC
Frequency: 50-60 Hz
Power Draw: I_{max} : 3.15 A
Overvoltage: Category III
Pollution: Degree 1

2.5 COMMUNICATIONS

Standard: (2) Standard RS485 / MODBUS Protocol
Optional: 4-20mA, Fiber Optic, HART & Ethernet

2.6 MECHANICAL

Explosion Group of Ex Atmosphere: IIC
Housing Mat: Stainless Steel
Dimensions: 125 cm x 76 cm x 37 cm
Weight: 180 kg
Ambient Temp: -40°C to +60°C
Safety Certs: IECEx
Protection class: Designed to meet IP65
Seismic Rating: Designed to withstand 3.0 g
Drive Chain: SS304
Mounting: 6" ANSI 150#RF Flange
Op. Pressure: 400 mbar
Stilling Well: Not Required
Isolation Valve: Required
Pinch Valve: Required

2.7 INTERNAL CONNECTIONS

All internal connectors have a flammability rating of V-0. Skipped designators have been removed from the design. Not all connections are used, and some may not be present on the assembly.

SI-7000 Controller Assembly

J1 Mains Power Input (Universal AC Input)
J2 Field Wiring Input (RS485)
J6 Reel Encoder Input
J8 Motor Power Output
J10 Probe Interface Module (PIM)
J14 JTAG Programing (manufacturer use only)
J22 Alternative 28 VDC – 48 VDC Power Input

Probe Interface Module (PIM)

J1 SI-7000 Controller Assembly
J2 Slip-ring
P1 Primary Earth Ground
P2 Optional Earth Ground

3 WARNINGS, CAUTIONS, AND NOTICES

Scientific Instruments, Inc. is not liable for any injuries, death, or damages caused during the operation of the SI-7000. It shall be the customer's responsibility to ensure all operations and maintenance are performed in a safe manner and complies with all government, local and governing agency regulations



WARNING: All applicable government, local, and governing agency regulations shall be strictly adhered to. Failure to do so could result in death or serious injury



WARNING: External power supply shall be disconnected via switch or circuit breaker prior to opening the electronics enclosure lid or removing the mechanical enclosure cover. Failure to do so could result in death, serious injury, or extensive property damage



WARNING: Care must be taken to ensure adequate clearances are maintained between structures, other equipment, and personnel when lifting and moving the SI-7000 into position. Failure to do so could result in death, serious injury, or extensive property damage



WARNING: Possibility of cryogenic burns! If the SI-7000 is installed on tanks with cryogenic liquids (such as LNG), when the probe is lifted for maintenance it must be given time to reach ambient temperatures to avoid cryogenic burns when handling the probe. It usually takes more than 4 hours to reach ambient temperature. When the temperature indicated on the touch screen main display reaches ambient temperature, it should be safe to open the mechanical unit.



WARNING: The SI-7000 is not for intended oxygen service



WARNING: Do not open when an explosive atmosphere is present.
Maximum experimental safe gap (MESG) of gas/air mixture $\geq 0,05$ mm



CAUTION - PINCH POINT: Care must be taken when closing the SI-7000 electronics enclosure lid and all personnel shall keep their hands clear. Failure to do so may result in injuries as fingers may be pinched.

SPECIAL CONDITIONS OF USE: Flameproof joints are not intended to be repaired.

NOTICE: There are no consumable materials associated with using the SI-7000.

NOTICE: During normal operation there are no surfaces which can cause burns due to high temperatures. If the equipment fails in a manner not covered in the service manual, please consult with Scientific Instruments before touching the equipment.

NOTICE: Instructions for cleaning the probe are given in the service manual. This is the only whetted part that may require periodic cleaning. The drive chain does not need to be cleaned, and no un-whetted parts require periodic cleaning. In the range of applications, the instrument is designed for, no decontamination is required.

NOTICE: All operation instructions in this manual shall be followed. Failure to do so may result in damage to the SI-7000 and void warranty.



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Part B

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4400 W. Tiffany Dr. West Palm Beach, FL 33407 U.S.A.
090-7000-2B Rev A

Record of Revisions						
Rev	RFEA #	Description	Rev by	Date	Appr by	Date
-		Initial Release	KN	8/21/19	KN	8/21/19
A		Updated Photos	JR	11/16/20	KN	11/16/20

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1 THEORY OF OPERATION

1.1 OVERVIEW

The SI-7000 controller receives input from the sensors in the mechanical unit and sends signals to control the motor depending on the current mode of operation. This controls the movement of a multi-sensor probe assembly suspended within the tank. Two level sensors inside the probe indicate to the controller whether the probe is in liquid, in vapor or at the interface between liquid and vapor. The controller monitors the signals from the temperature sensor and the density meter, and continuously calculates updated information for transmittal to a host computer.

The system has several modes of operation to accommodate maintenance activities and daily operation. The system may be programmed to perform a profile at regular intervals (as determined by customer requirements) to provide updated information about the conditions in the tank. When not performing a profile, the system is usually set to track liquid level, providing the most up-to-date reading of liquid level.

1.2 LIQUID LEVEL

The liquid level sensors produce different voltages depending on whether they are in liquid or vapor. The controller uses these differences in the sensor voltages to determine their state. By vertically spacing the level sensors a short distance apart, the controller can "recognize" liquid level at the point where the lower level sensor is "in liquid" and the upper level sensor is "in vapor".

1.3 TEMPERATURE

The temperature sensor is a four-leaded platinum element with a resistance of 100 Ω at 0 degrees centigrade. Using a one-milliamp excitation current, a sensor voltage is developed across the sensor proportional to its resistance. This voltage is digitized by an analog to digital converter and compared to data tables in the controller. The appropriate temperature is calculated and constantly updated as the temperature changes.

1.4 DENSITY

The density meter and its associated maintaining amplifier output to the controller a signal whose frequency is proportional to density. The controller determines the frequency of this signal and using a density conversion formula along with calibration parameters for the particular density meter, calculates the density of the medium into which the densitometer is submerged.

1.5 BOTTOM REFERENCE SWITCH

A mechanical switch assembly is employed to signal the controller when the probe is in contact with the tank floor. When the controller senses this "Probe at Reference" condition, it ceases downward movement of the probe and sets the probe position to the pre-programmed bottom reference value. The bottom reference value accounts for the distance of the level sensors from the bottom of the probe.

1.6 TOP REFERENCE SWITCH

A second mechanical switch assembly is employed to signal the controller when the probe has reached the top of the probe chamber. When the controller senses this "Probe at Top" condition, it ceases upward movement of the probe. This provides an easy method of parking the probe at the top in preparation for maintenance activities.

1.7 OPERATIONAL MODES

The system can be operated in any one of five principal modes using the touch screen interface pictured below. The available modes are Automatic (Auto), Calibration, Profile, Zoom Scan, and Manual. The most commonly used modes in daily operation are Auto and Profile. Manual is used normally only for maintenance activities. The Calibration mode can be used after maintenance activities. When power has been removed. It will quickly find bottom reference again and then return to Auto mode at the surface, or whenever it is desirable to re-establish the liquid level.



1.7.1 AUTOMATIC MODE (AUTO)

When the system is placed into Auto, the controller causes the probe to locate and track the liquid/vapor interface. The system is at the interface when the lower sensor is in liquid and the upper sensor is in vapor. The Automatic mode is the normal mode of operation; all alarms are reported, and profile runs can start automatically if they are programmed. The system returns to Auto upon completion of a profile run or a calibration run (described below).

1.7.2 CALIBRATION MODE (CAL)

When Cal mode is selected, the controller drives the probe to the bottom of the tank to establish bottom reference and then back to the liquid level interface, upon which the system is returned to Auto. A calibration is most often used after a power loss or after certain maintenance activities. When power is lost, the last probe position is stored, but it is assumed that the probe may have been moved manually while power was off (such as during maintenance activities), so on power-up the system is marked as not being calibrated.

1.7.3 PROFILE MODE

In the Profile mode, temperature and density readings are taken from the bottom of the tank up to liquid level, providing operators with an accurate representation of the current conditions in the tank.

When the Profile mode is started, the probe is driven to the tank floor where it re-establishes bottom reference. At the bottom, the probe pauses for a programmed delay time to allow conditions in the tank around the probe to stabilize. After the prescribed delay, position, temperature, and density readings are taken, and then the probe is driven up, stopping at programmed increments to collect temperature and density data. The same delay occurs at each point to allow readings to stabilize. The profile is terminated when the maximum number of points is taken or liquid level is found.

The parameters that control a profile may be entered via the LCD interface or over the Modbus serial interface.

After all readings are taken, the data is analyzed and alarms are generated as applicable. The system is returned to Auto after completion of the profile. In this manner, a complete profile of the tank is collected and stored for output to a host computer.

NOTE: The SI-7000 was designed to perform profiles when liquid is fully at rest and stabilized. Do NOT attempt to perform a profile while loading or unloading liquid in the tank. Loading LNG either top or bottom creates turbulence, swirls, and insatiability in liquid. High power compressors are used during unloading, this process also creates low level turbulence and instability in the liquid. For best

accuracy and results profiling must be scheduled a day before or after the loading or unloading process. Consult factory for more information.

1.7.4 ZOOM SCAN MODE

This feature is not implemented at this time.

1.7.5 MANUAL MODE

The Manual mode is used whenever it is desirable to take control of the probe and give specific commands to drive up or down. This would occur most often during maintenance activities. The probe may be stopped so that it will not track liquid level, or it may be driven up or down in fast, medium, or slow.

NOTE: No alarms are reported in the Manual mode and the liquid level reading is not updated. It is important to realize that the system will remain in the Manual mode indefinitely until it is returned to Auto by an operator or by maintenance personnel.

1.8 POWER UP CONDITION

The SI-7000 takes its reference from the floor of the tank; therefore, it is important to re-establish this bottom reference after power up. If this reference has not been established, the current probe position will be marked as "Uncal".

The system can be set so that it will automatically reestablish its bottom reference after a power loss, or it can be set so that it will power up in Manual with the motor stopped. This setting ("Change to Start in Auto" or "Change to Start in Manual") is found in the Configuration section accessed via the LCD interface. If it is set to start in Auto mode, it will first find the tank bottom and then return to liquid level.

If the system is not set to reestablish bottom reference automatically, it is only necessary to start a Cal run (enter Calibration Mode). The probe will travel to the bottom of the tank, re-establish bottom reference and then return to liquid level, where it will return to the Auto mode.

2 OPERATOR INTERFACE

Information about the system status is usually transmitted to a host computer (or DCS) in the Control Room using the Modbus protocol. The information available includes the current mode of operation, direction and speed of travel of the probe, current values for the position, liquid level, temperature, and density, profile data, and various status indications and alarms, but the programming and implementation of an optional DCS operator interface for the SI-7000 may be different at each site, therefore some of the features described below may not appear. The sections below describe what is available on the SI-7000 LCD screen.

2.1 HOME SCREEN

The home screen is the default display when the gauge is powered on. This screen displays the level, temperature, and density, along with other key parameters.



The current values for position, temperature, and density are the values measured where the probe is located at the moment. If the density is at a very low value (at or equal to zero), it is possible that the probe is in vapor instead of liquid.

The current value for liquid level should also be displayed if it has been established after power-up. If it has not yet been found, it will be zero or the last recorded value. Note that the liquid level can only be guaranteed to be correct when the probe is actually at liquid level. This is indicated by the "At Liquid Level" indicator as described below. In addition, to guarantee an accurate liquid level there should be no "Probe Uncal" status indication (also described below).

There are a number of status indicators that are very important to system operation and must be understood to get a complete picture of the current conditions.

The most basic indicators are the upper and lower level sensors, along with the bottom reference switch and top limit switch. The bottom switch is wired across the lower level sensor, and the top limit switch is wired across the upper sensor. Thus each sensor has 3 possible states:

- Closed: The corresponding switch is closed and the sensor is shorted, so it is not possible to indicate liquid or vapor.
- Liquid: The corresponding switch is open and the sensor is in liquid. (This will be indicated by L-LIQ for the lower sensor, and U-LIQ for the upper sensor.)

- Vapor: The corresponding switch is open and the sensor is in vapor. (This will be indicated by L-VAP for the lower sensor, and U-VAP for the upper sensor.)

When the probe is fully submerged, both switches will be in liquid. If the probe is completely out of liquid, both sensors should indicate vapor. If the probe is at the interface (liquid level), the lower sensor should be in liquid, and the upper sensor should be in vapor.

The SI-7000 measures level with respect to the bottom of the tank, so to report an accurate liquid level, it must first establish the location of tank bottom. If this reference has not been established, the current probe position will be marked as "Uncal". The system saves its last position when power is lost. However, since it is possible that maintenance personnel have physically moved the probe and its cable during a power loss, the position of the probe is always marked as "Uncal" when the system is first powered up. Bottom reference must be reestablished to clear this condition.

The "Probe Uncal" indicator will also come on if the probe position decreases below where bottom reference was found previously without finding bottom reference there again.

The SI-7000 provides a number of alarms to alert operators to abnormal or potentially dangerous conditions. Some of the alarms relate to current conditions and some relate to conditions found in the tank during a profile run.

2.1.1 REEL ALARM

If the Reel Alarm indicator is on, it indicates that the mechanical unit reel is not turning, and it could be indicative of a mechanical failure in the drive mechanism or an obstruction in the tank. Maintenance personnel should investigate.

2.1.2 PROBE AT LIQUID LEVEL

"Probe at Liquid Level" should be on when the system is in its Auto mode of operation and is at liquid level. If this indicator is not on, the operator cannot be assured that the current liquid level reading is correct. If the probe is beneath the surface or it is up in the vapor, there is no way of knowing if liquid level is changing.

2.1.3 LEVEL ALARMS

Probably the most important alarmed parameter relating to current conditions is level. There are several different level alarms: a low level alarm, a low low level alarm, a high level alarm, and a high high level alarm. Each alarm level setting should be determined by operations personnel and programmed accordingly.

2.1.4 TEMPERATURE AND DENSITY ALARMS

The other alarms relating to current conditions are the high and low temperature alarms and the high and low density alarms. In most environments, these alarms are not critical, since it is the deviation alarms that are normally used to detect abnormal variations of temperature and density in the profile of the tank, not the high and low alarms.

2.1.5 DEVIATION ALARMS

During a profile run, temperature and density data are collected at each programmed stop point. At the end of the run, the data points are analyzed. If the temperature deviation between one point and the next point is greater than the programmed alarm value, the temperature deviation alarm is set. Similarly, if the density variation from one point to the next is greater than the alarm threshold value, the density deviation alarm is set. These alarms are not cleared until another profile run is made, and it is verified that the alarmed condition no longer exists.

The deviation alarms are very important, since a variation in density or temperature could indicate layering or stratification, which if left unchecked could lead to a potential rollover condition.

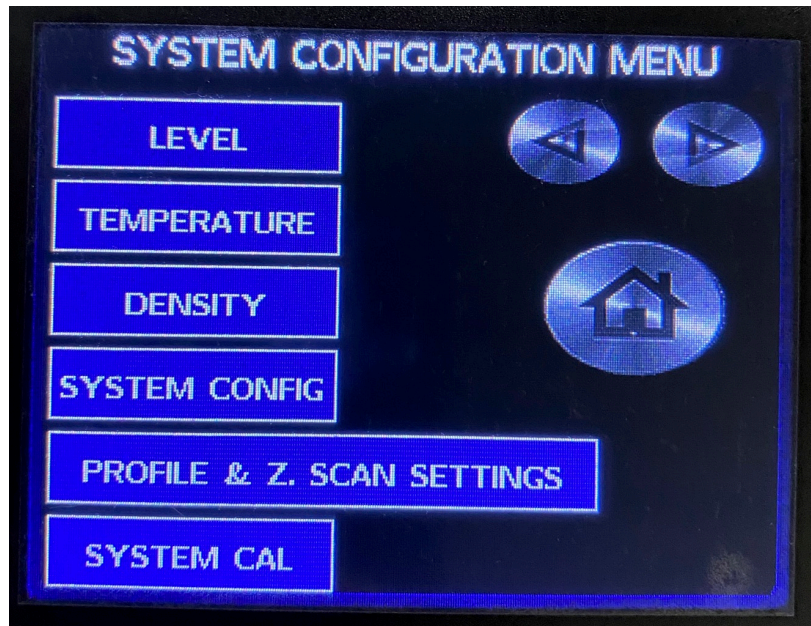
2.1.6 ALARM FUNCTIONALITY IN MANUAL MODE

NOTE: It is very important to understand that when the system is in its Manual Mode of operation, no alarms will be reported, even if alarm conditions exist. For the system to report alarms, it must be in the Auto mode. The system will immediately switch to Auto mode after a Profile or Cal run.

2.2 SYSTEM INFO

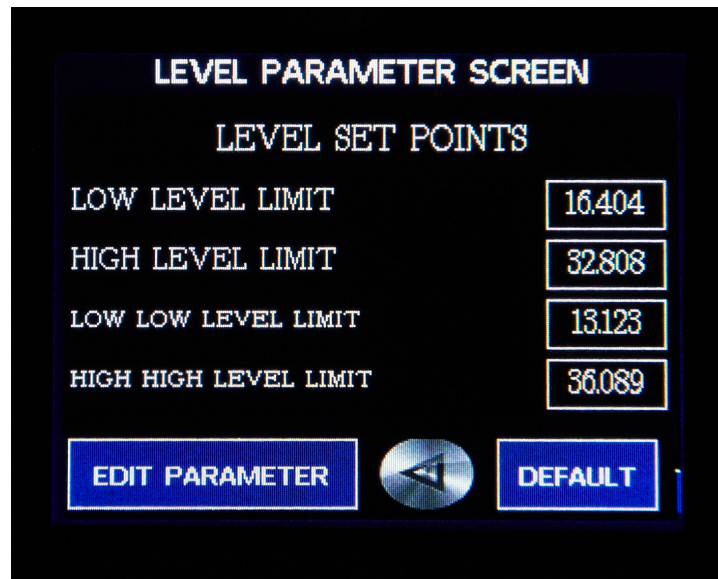
This screen displays the software revision and the Tank ID.

2.3 SYSTEM CONFIGURATION MENU



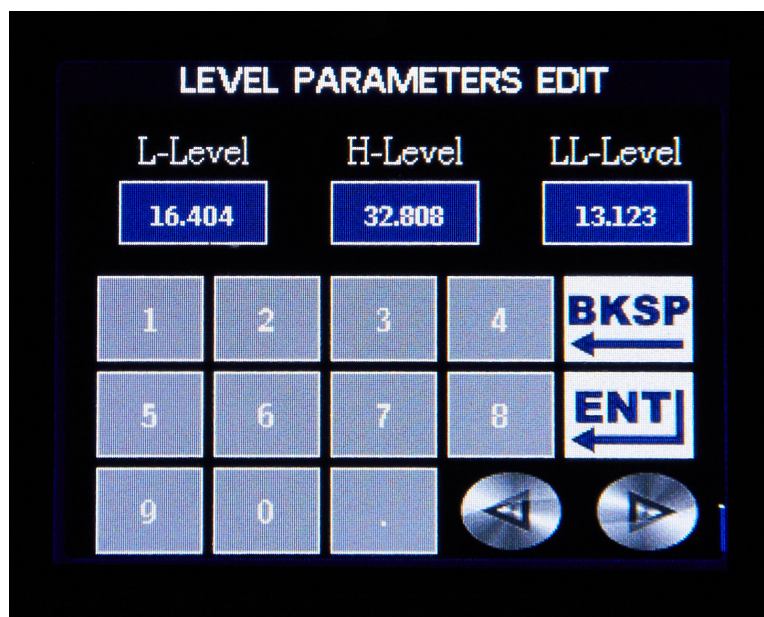
Pressing CONFIG on the main screen brings up the System Configuration Menu. This allows access to parameters that can be changed via the LCD touch screen display.

2.3.1 LEVEL



One of the primary functions of the LTD is to measure the level of the liquid in a storage tank. If the level falls below the low limits or rises above the high limits, alarms are generated. The set points for these alarms are set in this menu.

To change the values, push on the “Edit Parameter” button. A new screen will appear allowing the values to be changed.



To change the levels, touch the blue squares and then enter the desired reference. When all the levels are entered, click enter and save.

Use the right arrow to access the fourth parameter.

2.3.2 TEMPERATURE

TEMPERATURE PARAMETERS

N CAL. POINT	ICE CAL. POINT
22.000	100.000

ALARM SET POINTS

TEMP. LOW ALARM LIMIT	-238.0000
TEMP. HIGH ALARM LIMIT	-274.0000
TEMP. DEVIATION ALARM LIMIT	0.7200

EDIT PARAMETER  DEFAULT

The temperature sensor in the SI-7000 is mounted inside the density meter. Each sensor is theoretically a 100 Ω platinum resistance thermometer (PRT), but the resistance can vary slightly from the theoretical value. Before each one is shipped, a calibration process is performed, and the exact value is measure in ice and nitrogen. The resulting values are included on the calibration certificate for each density meter.

The SI-7000 electronic circuits are calibrated to measure resistance accurately, so once the correct values are entered for the sensor being used, the instrument can display temperature within the tolerance stated in the system specification.

Temperature Measurement Range: -200°C to +50°C

Error (over-range): 999.99°C

Error (under-range): -555.55°C

Low Temperature and High Temperature alarms can be triggered at set points determined by the customer and are entered in Kelvin.

The Deviation alarm is a very important alarm that is used to determine if the tank is stratified or not. In a profile, if there is a difference from one point to the next greater than the Deviation alarm limit, a Temperature Deviation (TD) alarm is triggered. This alarm will remain active until another profile is performed and the condition is cleared.

2.3.3 DENSITY

DENSITY PARAMETERS

T0	D0	K
187.000	190.000	1.000

ALARM SET POINTS

DENSITY LOW ALARM LIMIT	24.971
DENSITY HIGH ALARM LIMIT	31.214
DENSITY DEVIATION ALARM LIMIT	0.050

EDIT PARAMETER  **DEFAULT**

The density meter is a calibrated device. Before each one is shipped, a calibration process is performed, and three coefficients are determined and included on the calibration certificate for each density meter.

The SI-7000 electronic circuits are designed to measure frequency accurately, so once the correct values are entered for the density meter, the instrument can display density within the tolerance stated in the specification.

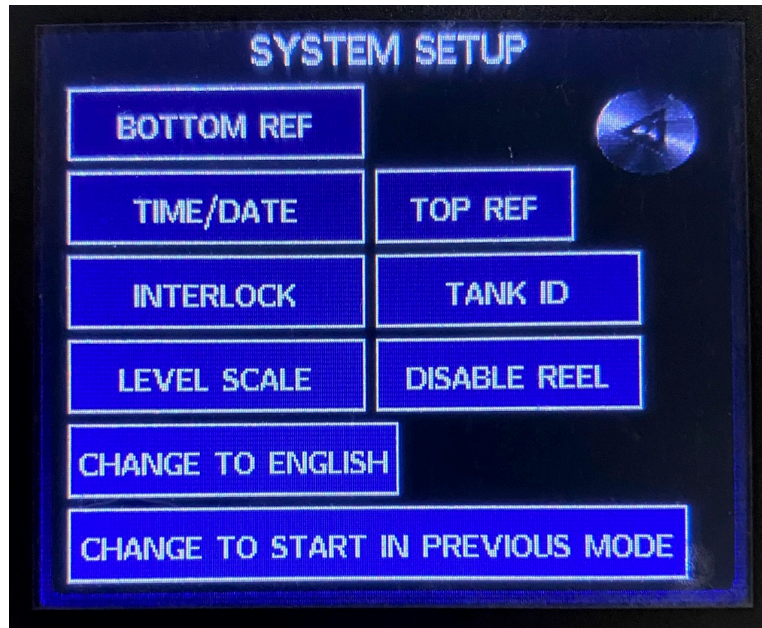
Density Measurement Range: 400 kg/m³ to 600 kg/m³

Error (over-range): 9999.99 kg/m³

Low Density and High Density alarms can be triggered at set points determined by the customer and are entered in kg/m³.

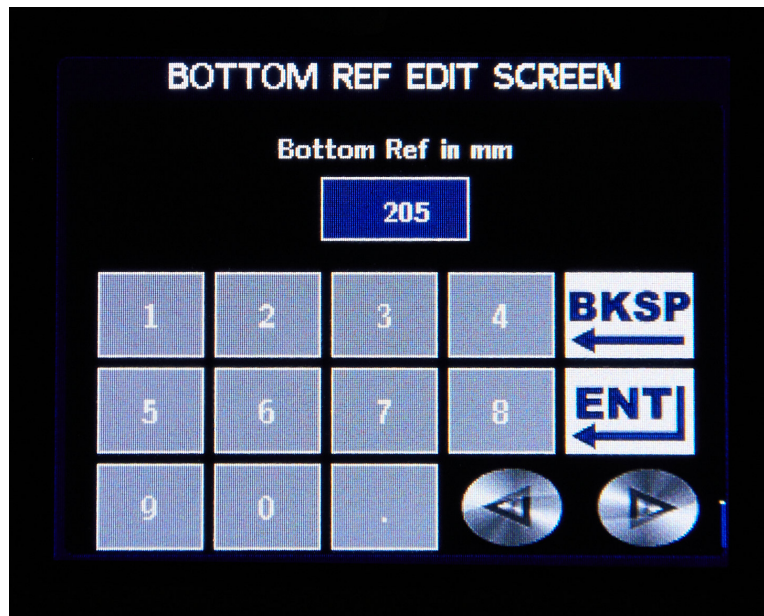
The Deviation alarm is a very important alarm that is used to determine if the tank is stratified or not. In a profile, if there is a difference from one point to the next greater than the Deviation alarm limit, a Density Deviation (DD) alarm is triggered. This alarm will remain active until another profile is performed and the condition is cleared.

2.3.4 SYSTEM



The System Menu provides access to a number of sub menus.

2.3.4.1 BOTTOM REFERENCE



The SI-7000 measures level with respect to the bottom of the tank, and it touches bottom at the beginning of every Cal run and Profile. When the probe touches tank bottom, the motor pulse count is set to zero.

The height of the lower level sensor above the bottom of the probe is added to the pulse count to establish an accurate position of the lower level sensor.

To change the bottom reference, touch the blue square and then enter the desire reference. Click enter and save.

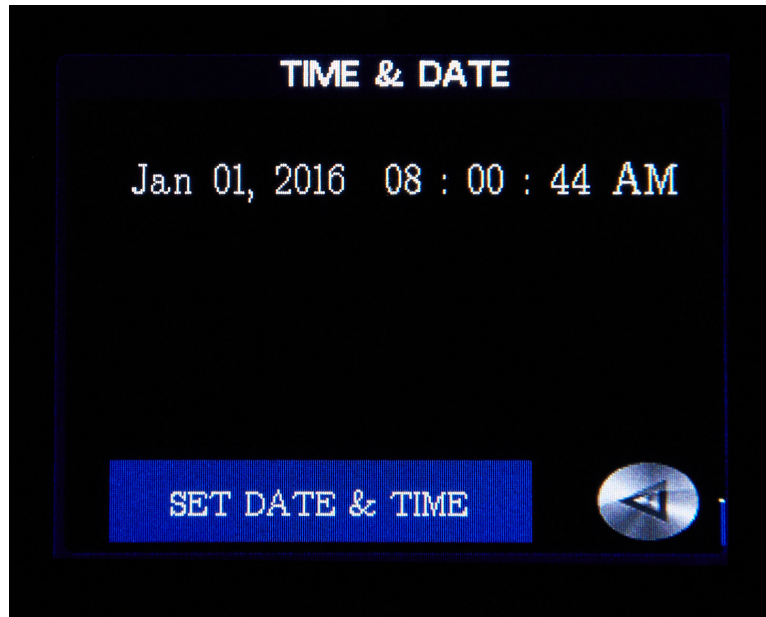
2.3.4.2 TANK ID



For installations with multiple tanks, the Tank ID distinguishes one tank from another. This is also the Modbus Address.

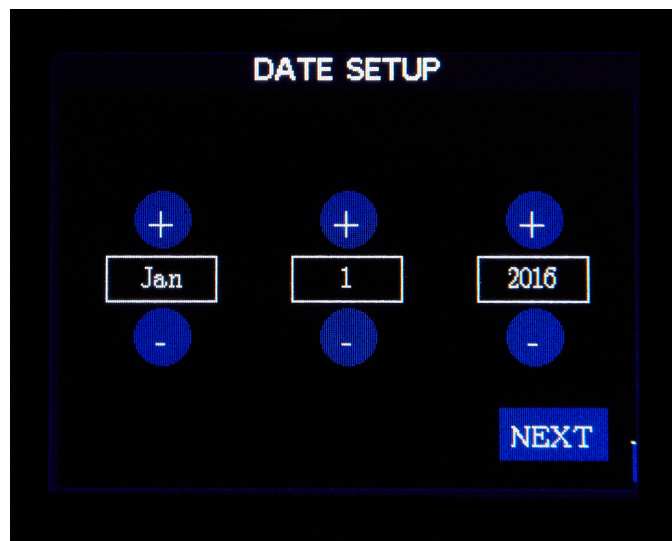
To change the Tank ID "Address", touch the blue square and then enter the desire ID "Address". Click enter and save.

2.3.4.3 TIME/DATE



The SI-7000 maintains a record of the date and time and keeps the time for up to one month after power is turned off. If the controller is powered off for more than one month, it may be necessary to reprogram the current date and time.

To change the date and time, push the button with that label. A new screen opens with an option to set the date.



After specifying the correct date, push Next, and another screen opens with an option to set the current time. Push Done when finished.

2.3.4.4 LEVEL SENSOR CAL CLOSED

This is for internal use, and a password is required.

2.3.4.5 MICROCONTROLLER CAL

This is for internal use, and a password is required.

2.3.4.6 LEVEL SENSOR CAL LIQUID

This is for internal use, and a password is required.

2.3.4.7 UNITS OF MEASURE (METRIC/ENGLISH)

This button gives the option to change to the other system of measure. If the system is currently in metric, the button will display an option to Change to English. If the system is in English units, the button will display an option to change to metric.

2.3.4.8 100 OHM CAL

When this menu is selected, a verification message is displayed.

“Calibrate at 100 OHM?”

This is for internal use, and should only be performed when instructed to do so by the technical support department. Press “No” to exit the menu without calibrating.

2.3.4.9 NITROGEN TEMPERATURE CAL

When this menu is selected, a verification message is displayed.

“Calibrate at Nitrogen temperature?”

This is for internal use, and should only be performed when instructed to do so by the technical support department. Press “No” to exit the menu without calibrating.

2.3.4.10 LEVEL SCALE

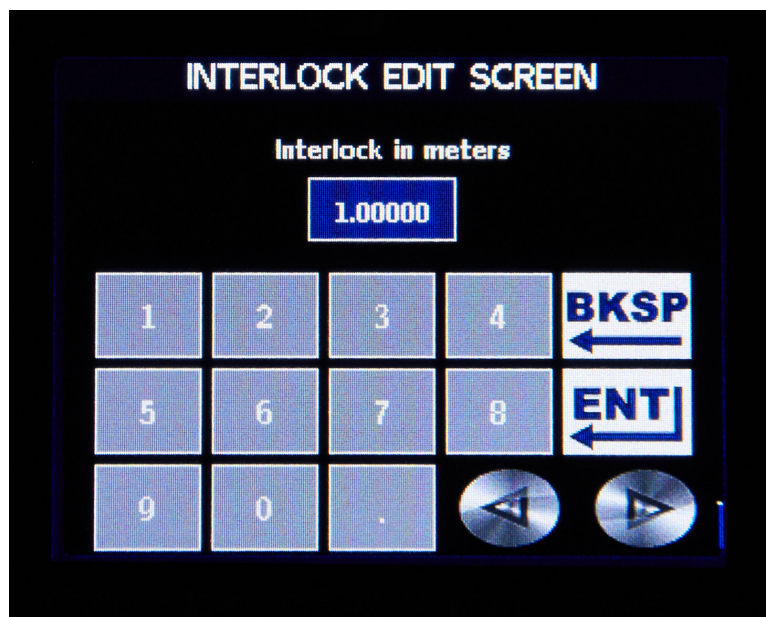


The formula for level is as follows:

$$\text{Level} = \text{Motor Pulses} * \text{Level Scale} + \text{Bottom Reference}$$

So Level Scale is the calibration value that relates pulses to actual probe position. A calibration is done at the factory, and Level Scale value is determined and then recorded on a site specific Configuration Worksheet. The value shown here should match the value in that Configuration Worksheet.

2.3.4.11 INTERLOCK



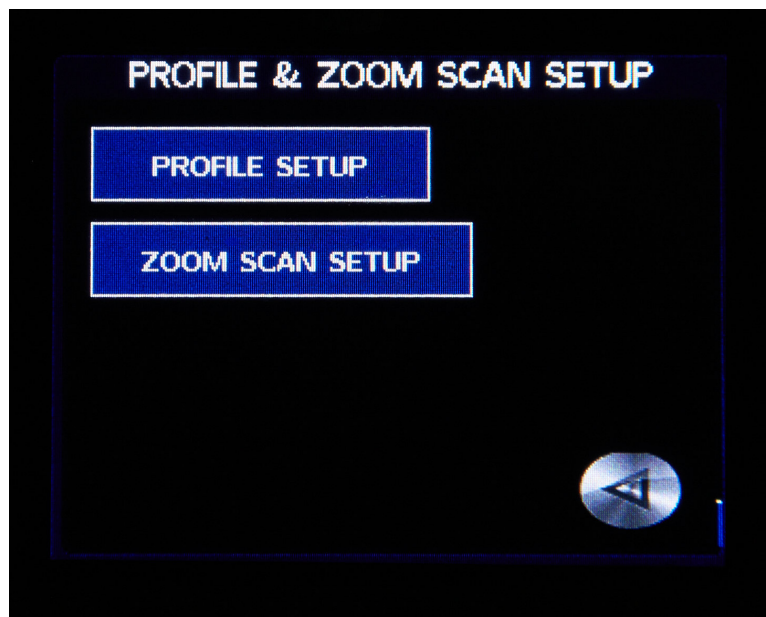
Interlock is the high limit for probe movement. It is normally set to keep the probe down in the cold part of the tank. If the Interlock indicator in the DCS or PC User Interface is on, it means that the probe has reached the highest position allowed. Unless the probe was raised to that point manually, this is an abnormal situation and maintenance personnel should investigate.

To change the interlock, touch the blue square and then enter the desire interlock. Click enter and save.

2.3.5 COM SETTINGS

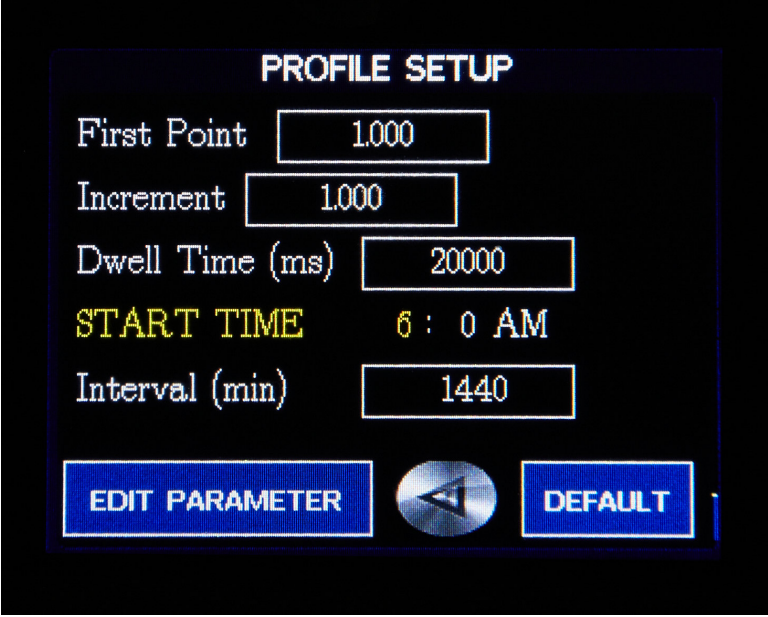
This menu is not supported at this time.

2.3.6 PROFILE & ZOOM SCAN SETTINGS



This menu has two sub menus.

2.3.6.1 PROFILE SETUP



PROFILE SETUP

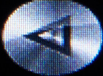
First Point 1.000

Increment 1.000

Dwell Time (ms) 20000

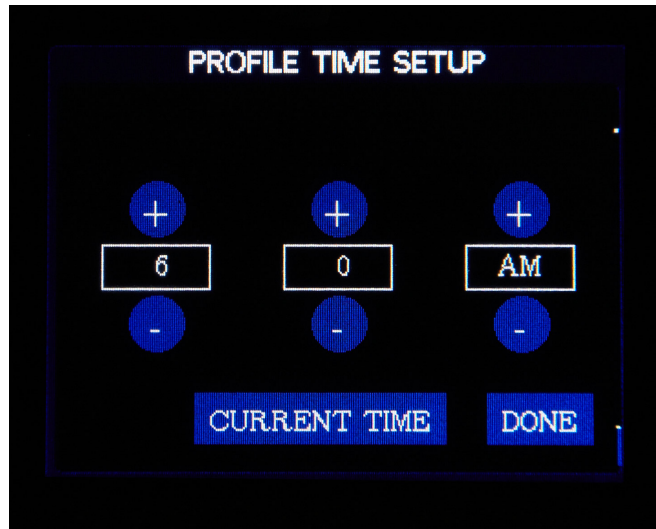
START TIME 6 : 0 AM

Interval (min) 1440

EDIT PARAMETER  DEFAULT

One of the most important features of the SI-7000 is the ability to collect data at various points in the tank. A profile starts at the bottom of the tank and points are taken in the upward direction until the maximum number of points is reached or liquid level is found. This menu allows customization of the profile that is taken.

The actual first point of the profile is taken at the tank bottom. The first programmable point is specified in the field, "First Point." As the probe moves off the bottom, it will first stop at this point, and then the Increment will be added to calculate the next stop point. The Dwell Time is the time in milliseconds that the probe will wait at each point before taking readings. The Interval is the amount of time in minutes before the next profile will be taken.



The time setup screen makes it possible to choose the exact time when the profile will start. Note that the profile run will not start if the system is in Manual when the programmed time arrives, but will start if the system is in Auto, Calibrate, or Profile.

2.3.6.2 ZOOM SCAN SETUP

This menu is not implemented at this time.

2.3.7 START MODE (AUTO/MANUAL)

When the system is powered on, the power up state is programmable. It can either remain in Manual Mode, waiting for instructions, or it can go directly into Auto Mode and measure liquid level. During maintenance, it is usually set to stay in Manual. In normal operation it can be changed to start in Auto.

If it is currently set to start in Manual, the button will show the option to change to start in Auto. If it is set to start in Auto, the button will show the option to start in Manual.

2.4 COMMAND MENU



The Command Menu allows the current LTD mode to be changed.

Pressing **Manual Mode** will cause the probe to go to Manual Stop. It will remain in this mode indefinitely, waiting for operator commands.

If you push the **Level Calibration** button, the probe will move to the tank bottom to re-establish the bottom reference, and then it will return to the liquid level.

The **Auto Mode** button will return the LTD to normal Auto mode, where it continuously tracks liquid level.

Push **Start Profile** to initiate a new profile, moving to the bottom of the tank, and then back to liquid level, collecting data at the programmed stop points.

Zoom Scan is not implemented at this time.

Use the Back button to return to the previous menu.

2.4.1 MANUAL MODE

In Manual mode the probe is under operator control. Pressing the Up arrow one time starts the probe moving up slowly. Press it again to change to medium speed and press it a 3rd time to change to high speed.

Likewise, pressing down one, two, or three times puts it in slow, medium, or fast speed moving down.

The Stop button should be pressed before changing directions to avoid jerking the chain and probe.

2.4.2 LEVEL CALIBRATION

Pressing Level Calibration starts the probe moving down to the tank bottom. After it reaches the tank bottom it will return to liquid level. After the calibration button is pushed, the display returns to the normal display.

2.4.3 AUTO MODE

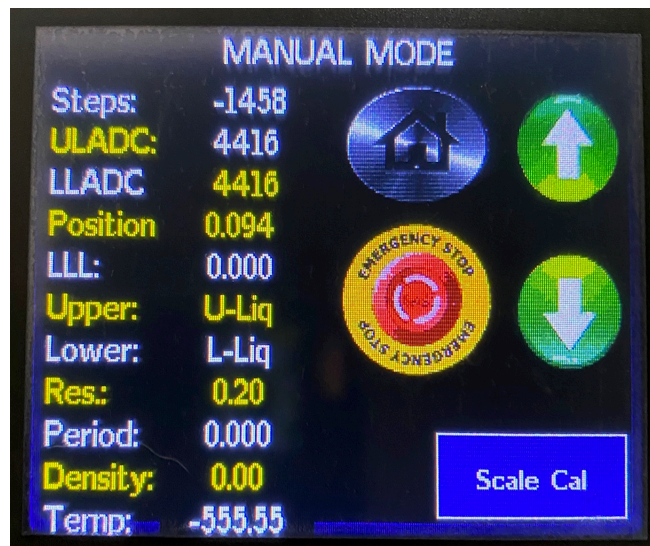
Pressing Auto Mode puts the LTD in Auto, and the probe will return to liquid level from wherever it is, and it will track liquid level continuously. After the Auto Mode button is pushed, the display returns to the normal display.

While in Auto, the probe usually drives in slow to track liquid level, but it can accelerate to a faster speed as needed.

2.4.4 START PROFILE

Pressing Start Profile starts the probe moving down to the tank bottom. After it reaches the tank bottom and collects data at the first point, it begins moving up, stopping to collect data at the programmed stop points along the way. When it reaches liquid level, the last point will be taken, and the profile will complete. After the Start Profile button is pushed, the display returns to the normal display.

2.5 MOTOR



Pressing MOTOR brings up a screen that shows detailed diagnostic information related to the motor drive function and other system parameters.

3 TECHNICAL ASSISTANCE

For technical assistance, please contact Scientific Instruments via telephone, fax, mail, or email.

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4 DISCLAIMER

WARNING: Only personnel **certified by Scientific Instruments, Inc.** are authorized to execute the activity of installation, commissioning, or any type of work/maintenance on the model SI-7000 LTD gauge. Unauthorized activity on the SI-7000 LTD gauge will negate warranty. Scientific Instruments, Inc. assumes no responsibility for injury, loss or damage to person or property caused by unauthorized activity. Any questions please contact us at email fieldservicetechns@scientificinstruments.com or Phone: (800)466-6031.